

TOL'KA, Russian Federation

WMO: 236620

Lat: 63.9944N

Lon: 82.0475E

Elev: 114

StdP: 14.64

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.6	-43.4	-54.5	0.2	-49.9	-48.6	0.3	-43.5	13.9	4.2	12.1	4.5	1.3	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	84.7	64.8	80.4	63.0	76.0	61.6	66.8	79.1	64.9	76.6	63.0	73.7	5.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	85.1	71.4	60.5	79.0	68.8	58.6	73.9	66.9	31.5	78.7	30.0	76.9	28.6	73.4	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.6	10.0	DB	-51.1	89.0	5.0	3.1	-54.7	91.2	-57.7	93.0	-60.5	94.8	-64.1	97.0	(4)
(5)			WB	-51.1	69.3	5.0	2.1	-54.7	70.9	-57.7	72.1	-60.5	73.3	-64.1	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	-13.6	-7.9	11.8	24.5	35.1	57.3	61.0	53.9	43.4	26.6	0.4	-6.8
	DBStd	28.95	16.31	16.76	14.16	12.14	10.06	10.07	8.15	6.78	8.08	12.10	17.59	19.56
	HDD50	10315	1973	1620	1183	766	477	45	7	36	231	728	1488	1761
	HDD65	15064	2438	2040	1648	1215	927	269	178	346	648	1191	1938	2226
	CDD50	819	0	0	0	0	15	264	347	158	33	2	0	0
	CDD65	95	0	0	0	0	1	38	53	3	0	0	0	0
	CDH74	1299	0	0	0	0	13	514	722	50	0	0	0	0
	CDH80	366	0	0	0	0	4	137	225	0	0	0	0	0
Wind	WSAvg	4.8	3.9	4.1	5.3	5.7	6.0	5.2	4.5	4.4	4.8	5.1	4.4	4.0
Precipitation	PrecAvg	21.7	1.1	0.9	1.0	1.3	1.7	2.2	2.7	3.1	2.2	2.5	1.7	1.4
	PrecMax	29.5	2.0	1.9	2.4	3.0	3.5	4.4	5.6	6.1	4.9	4.7	3.0	2.3
	PrecMin	13.9	0.2	0.4	0.2	0.4	0.6	0.7	0.3	1.0	0.7	0.7	0.7	0.9
	PrecStd	3.9	0.5	0.4	0.5	0.6	0.7	1.0	1.4	1.3	0.9	0.9	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	31.7	39.6	55.8	73.8	88.3	89.7	78.6	71.8	55.6	37.4	31.3
		MCWB	20.7	30.1	32.6	43.6	55.8	66.1	66.8	62.9	58.9	49.3	36.2	30.1
	2%	DB	17.2	23.9	36.1	48.8	65.2	84.4	85.8	75.0	65.4	48.7	33.2	28.1
		MCWB	16.3	22.4	31.3	39.2	52.6	64.6	65.2	62.0	56.2	45.4	31.9	27.2
	5%	DB	13.8	20.4	33.5	44.5	57.5	80.3	82.1	70.9	60.4	44.2	30.1	24.3
		MCWB	13.1	19.3	30.1	37.3	48.7	62.3	64.1	60.6	53.9	41.8	29.1	23.4
	10%	DB	9.8	15.8	31.0	40.6	50.8	76.0	77.7	66.6	55.8	40.9	26.5	19.7
		MCWB	9.0	14.8	28.3	35.4	42.9	60.7	62.8	58.9	50.6	38.7	25.4	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	30.1	33.5	43.3	57.1	68.8	70.5	67.0	60.7	49.9	36.3	30.2
		MCDB	20.9	31.4	37.0	52.0	68.5	83.2	83.7	73.7	69.2	53.8	37.4	31.2
	2%	WB	16.5	22.4	32.0	40.1	53.7	66.2	67.6	64.1	57.2	45.7	32.3	27.3
		MCDB	17.4	23.5	35.2	46.9	63.2	78.6	81.0	71.2	63.3	48.0	33.3	28.2
	5%	WB	13.0	19.3	30.4	38.2	49.0	64.4	65.6	61.6	54.4	42.4	29.1	23.4
		MCDB	13.7	20.5	33.2	43.7	58.0	77.8	78.2	67.9	59.7	44.1	30.0	24.3
	10%	WB	9.0	14.8	28.4	36.0	43.5	62.3	63.7	59.5	51.1	38.8	25.4	19.1
		MCDB	9.7	15.8	30.8	40.3	49.8	74.0	74.9	65.8	55.2	40.6	26.5	19.8
Mean Daily Temperature Range	5% DB	MDBR	14.1	17.6	19.1	18.6	15.3	20.3	19.7	17.0	13.4	10.1	13.4	13.7
		MCDBR	14.3	15.0	15.4	18.6	27.0	27.6	27.8	24.6	20.1	12.0	11.3	11.5
		MCWBR	13.8	14.5	11.6	12.0	14.5	11.9	11.1	12.9	11.5	9.0	11.4	11.4
	5% WB	MCDBR	14.5	14.7	14.4	15.6	26.5	24.1	24.0	19.5	17.2	11.5	10.7	11.5
		MCWBR	14.1	14.2	11.5	10.6	14.5	11.7	11.1	11.8	10.6	8.9	10.9	11.4
Clear-Sky Solar Irradiance	taub	0.227	0.280	0.328	0.349	0.360	0.377	0.430	0.402	0.349	0.297	0.218	0.224	
	taud	1.893	2.014	1.989	2.046	2.245	2.394	2.218	2.278	2.382	2.213	1.920	1.876	
	Ebn at Noon	98	192	230	257	268	266	245	241	232	188	99	33	
	Edh at Noon	10	22	35	41	37	33	38	33	24	18	9	4	
All-Sky Solar Radiation	RadAvg	60	300	791	1399	1679	1749	1567	1068	626	239	88	24	
	RadStd	7	27	53	142	153	142	171	123	93	38	14	2	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page